

CS4051 — Information Retrieval

Chapter 13: Text Classification — Complete Study Notes

Week 13 | Instructor: Muhammad Rafi | April 2026

These notes cover all topics discussed in class, including student questions and clarifications. Topics covered: Ad Hoc vs Standing Query, Text Classification, Naive Bayes, Multinomial NB, Bernoulli Model, Evaluation, Feature Selection (MI, Chi-Square, Frequency-Based).

Topic 1: Ad Hoc vs. Standing Query

Ad Hoc Query

A one-time information need. The user types a query, gets results, and the interaction ends. This is the traditional IR model covered in earlier chapters.

Standing Query

A query that persists over time. The system continuously monitors an incrementally growing document collection and categorizes new documents automatically as they arrive. Example: 'Notify me whenever something about AI is published.'

Why This Matters

Standing queries ARE classifiers — the system must decide for every new document: does this belong to a certain category? This is the motivation for automated text classification.

Q: When a document comes in, the system should classify its category and place it accordingly?

A: Yes, exactly! Like Gmail — when an email arrives it is immediately classified as Primary, Spam, Promotions, or Social without you doing anything. A classifier runs in the background at all times. Ad Hoc: User comes to system, asks a question, gets answer. Standing/Classification: System watches incoming docs and proactively categorizes them.

Topic 2: Text / Document Classification

Definition

Text classification is the process of assigning a predefined set of classes to a new incoming document.

Input	A document d + a fixed set of classes $C = \{c_1, c_2, \dots, c_J\}$
Output	A predicted class c_i from C

Four Types of Classification

Type	Description
Manual	Human experts write hand-coded rules. High accuracy if carefully refined but expensive.
Supervised	Learns from labeled training data (d, c) pairs. Most widely used approach.

Semi-supervised	Starts supervised then continues learning from its own classifications over time.
Unsupervised	Learns categories entirely on its own with no labels — discovers structure implicitly.

Classification Configurations

# Outputs	Output Type	Kind	Example
1 per instance	Binary	Binary	Spam / Not Spam
1 per instance	Multivalued	Multiclass	Sports / Politics / Tech
n per instance	Binary	Multilabel	Post tagged #food AND #travel
n per instance	Multivalued	Multidimensional	Complex multi-dimension tagging
1 per n instances	Binary/Multivalued	Multiinstance	Bag of documents, one label

Topic 3: Naive Bayes Classification

What Is It?

The first supervised learning method in Chapter 13. A probabilistic classifier — instead of hard rules, it calculates probabilities and picks the most likely class. It is called 'Naive' because it assumes all words in a document are independent of each other.

The Main Formula (Built Step by Step)

Step 1 — What we want (find the class with highest probability given document words):

$$\hat{y} = \operatorname{argmax} P(y \mid x_1, x_2, \dots, x_n)$$

Step 2 — Apply Bayes Theorem (flip it so we can compute it):

$$= \operatorname{argmax} P(y) \times P(x_1, x_2, \dots, x_n \mid y)$$

Step 3 — Apply Naive Assumption (words are independent, so multiply individually):

$$= \operatorname{argmax} P(y) \times \operatorname{PRODUCT} \text{ of } P(x_i \mid y) \text{ for } i = 1 \text{ to } n$$

$x_1, x_2 \dots x_n$ = words of the document | y = class being tested | $\hat{y}$ = predicted class Best Class = Prior x (word1_prob x word2_prob x word3_prob ...)

Q: In the formula, x are words of the document and y is the class?

A: Yes, exactly right! The formula asks: for each possible class y , multiply the probability of that class existing WITH the probability of each word appearing in that class. Whichever class gives the highest number — that is our predicted answer $\hat{y}$.

Probability Estimation — Maximum Likelihood

1. Prior $P(c_j)$ — How common is this class?

$$P(c_j) = \operatorname{doccount}(C = c_j) / N_{\text{doc}}$$

2. Likelihood $P(w_i|c_j)$ — How common is this word in this class?

$$P(w_i|c_j) = \operatorname{count}(w_i, c_j) / \operatorname{SUM} \text{ of } \operatorname{count}(w, c_j) \text{ for all } w \text{ in Vocabulary}$$

3. With Laplace Smoothing +1 — Avoids Zero Probability Problem:

$$P(w_i|c) = (\text{count}(w_i, c) + 1) / (\text{SUM count}(w, c) + |V|)$$

Why +1? If a word never appeared in training, its probability = 0, which kills the entire multiplication! Adding +1 to numerator and |V| (vocabulary size) to denominator fixes this.

Topic 4: Multinomial NB — Training and Worked Example

Training Data

Doc	Words	Class
1	Chinese Beijing Chinese	c (China)
2	Chinese Chinese Shanghai	c (China)
3	Chinese Macao	c (China)
4	Tokyo Japan Chinese	j (Japan)
5 (TEST)	Chinese Chinese Chinese Tokyo Japan	? — to classify

Vocabulary = {Chinese, Beijing, Shanghai, Macao, Tokyo, Japan} → |V| = 6

Step 1 — Priors:

$$P(c) = 3/4 = 0.75 \quad | \quad P(j) = 1/4 = 0.25$$

Step 2 — Likelihoods with +1 Smoothing:

Word	Class China (8 words total)	Class Japan (3 words total)
Chinese	$(5+1)/(8+6) = 6/14 = 3/7$	$(1+1)/(3+6) = 2/9$
Tokyo	$(0+1)/(8+6) = 1/14$	$(1+1)/(3+6) = 2/9$
Japan	$(0+1)/(8+6) = 1/14$	$(1+1)/(3+6) = 2/9$
Beijing	$(1+1)/(8+6) = 2/14 = 1/7$	$(0+1)/(3+6) = 1/9$
Shanghai	$(1+1)/(8+6) = 2/14 = 1/7$	$(0+1)/(3+6) = 1/9$
Macao	$(1+1)/(8+6) = 2/14 = 1/7$	$(0+1)/(3+6) = 1/9$

Step 3 — Classify Doc 5 (Chinese Chinese Chinese Tokyo Japan):

$$P(c|d5) = P(c) \times P(\text{Chinese}|c)^3 \times P(\text{Tokyo}|c) \times P(\text{Japan}|c) = 3/4 \times (3/7)^3 \times (1/14) \times (1/14) = 0.75 \times 0.0787 \times 0.0714 \times 0.0714 \approx 0.0003$$

$$P(j|d5) = P(j) \times P(\text{Chinese}|j)^3 \times P(\text{Tokyo}|j) \times P(\text{Japan}|j) = 1/4 \times (2/9)^3 \times (2/9) \times (2/9) = 0.25 \times 0.0110 \times 0.2222 \times 0.2222 \approx 0.0001$$

Result: $P(c|d5) = 0.0003 > P(j|d5) = 0.0001$ → Document 5 classified as CHINA!

Q: We took power of 3 on Chinese because it appears 3 times in the test document?

A: Yes, exactly right! 'Chinese' appears 3 times in Doc 5 so we write $(3/7)^3$. That is shorthand for $P(\text{Chinese}|c) \times P(\text{Chinese}|c) \times P(\text{Chinese}|c)$. Every word occurrence is multiplied separately — repeated words get their probability multiplied that many times!

Topic 5: Bernoulli Model

Key Difference from Multinomial

In Bernoulli, each word in the entire vocabulary gets a simple YES/NO (1 or 0): did this word appear in the document or not? Word frequency/repetition is completely ignored. Crucially, words that do NOT appear ALSO contribute to the probability — unlike Multinomial which simply ignores absent words.

	Multinomial	Bernoulli
Cares about	How many times word appears	Whether word appears or NOT
Word repetition	Counts multiple occurrences	Ignores — just 1 or 0
Absent words	Ignores completely	Also contribute to probability
Best for	Longer documents, more features	Shorter documents, fewer features
Parameter estimate	$P(X=t c)$	$P(U=1 c)$

Same Example — Bernoulli Result:

$P(c|d5) \approx 0.005$ | $P(j|d5) \approx 0.022$ Result: Bernoulli classifies Doc 5 as JAPAN — different from Multinomial! This shows that model choice matters depending on your data and document length.

Topic 6: Evaluation of Classification

Confusion Matrix — Binary (Spam/Ham)

	Actual Spam	Actual Ham
Predicted Spam	35 (True Positive) ✓	16 (False Positive) ✗
Predicted Ham	15 (False Negative) ✗	54 (True Negative) ✓

Three Key Measures

Measure	Formula	Divides by	Question it answers
Accuracy	$(TP+TN) / \text{Total}$	All documents	How many did I get right overall?
Precision	Diagonal / ROW total	Predicted class row	Of what I predicted as X, how many were really X?
Recall	Diagonal / COLUMN total	Actual class column	Of all real X documents, how many did I find?

Memory Trick: Precision = Predicted = Row (PPR) | Recall = Real/Actual = Column (RRC)

Q: Didn't we divide by total of ROW in recall and COLUMN in precision?

A: You are absolutely right — Precision uses ROW total and Recall uses COLUMN total. Precision C3 (ROW): $20 / (0+2+20+2+2) = 20/26 = 0.77$ Recall C3 (COLUMN): $20 / (0+1+20+12+0) = 20/33 = 0.61$

Multiclass Confusion Matrix

	C1	C2	C3	C4	C5
Pred C1	10	1	0	1	0
Pred C2	2	22	1	1	1

Pred C3	0	2	20	2	2
Pred C4	2	0	12	21	0
Pred C5	0	0	0	0	25

Green diagonal = correct predictions | Red = notable error (C4 confused 12 C3 docs as C4) Precision C1 (ROW): $10/(10+1+0+1+0) = 10/12 = 0.83$ Recall C1 (COLUMN): $10/(10+2+0+2+0) = 10/14 = 0.71$

Macro vs Micro Averaging

	Macroaveraging	Microaveraging
How	Calculate per class separately, then average	Pool ALL decisions into one table, then calculate
Treats classes	Equally regardless of size	By size — bigger classes dominate
Example result	$(0.5 + 0.9) / 2 = 0.70$	$100/120 = 0.83$
Use when	All classes equally important	Overall system performance matters

Q: I cannot understand micro and macro averaging with example?

A: Using the slide example (Class 1 and Class 2): MACROAVERAGING: Class 1 Precision = $10 / (10+10) = 10/20 = 0.5$ Class 2 Precision = $90 / (90+10) = 90/100 = 0.9$ Macro Average = $(0.5 + 0.9) / 2 = 0.70$ MICROAVERAGING: Pool together: TP = $10+90 = 100$, FP = $10+10 = 20$ Micro Precision = $100 / (100+20) = 100/120 = 0.83$ Real world analogy: For rare disease vs common cold: Macro = gives equal importance to rare disease and common cold Micro = common cold dominates because it has far more cases

Q: Precision of Class 1 would be $10/(10+10)$?

A: Yes exactly! Precision = diagonal / row total. Row of Class 1 = $10+10 = 20$, so Precision = $10/20 = 0.5$. You were confirming the calculation — which is correct!

Topic 7: Feature Selection

The Problem & Definition

Text documents can have thousands of words. Using ALL of them makes the model slow, adds noise, and hurts accuracy. Feature selection is the process of selecting only the most relevant and non-redundant subset of features for model training.

Benefits: (1) Reduces dimensionality (2) Simpler model, faster training (3) Improves accuracy by eliminating noisy features

Three Categories

Method	What it does	Speed	Overfitting	For Text?
Filter	Scoring metric independent of classifier	Fastest	Most prone	Most widely used
Wrapper	Uses classifier accuracy to evaluate each feature subset	Slow	Less prone	Not suitable
Embedded	Feature selection built into the model training process	Medium	Less prone	Good

Why Wrappers are NOT suitable for text: They must retrain the classifier for EVERY possible feature subset — with thousands of words this is computationally impossible.

How Feature Selection Works

For each term t and class c , compute a utility score $A(t,c)$. Keep only the TOP K terms with highest scores. This replaces a complex classifier (all features) with a simpler one (best subset only).

Topic 8: Mutual Information (MI)

What Is It?

Mutual Information measures how much information the presence OR absence of a term contributes to making the correct classification decision for a class. A high MI score means the term is very informative for that class.

The Four Counts Used (Contingency Table)

Notation	Meaning
N_{11}	Documents where term t IS present AND class c IS present
N_{10}	Documents where term t IS present BUT class c is NOT present
N_{01}	Documents where term t is NOT present BUT class c IS present
N_{00}	Documents where term t is NOT present AND class c is NOT present
N	Total number of documents

MI Formula:

$$I(U;C) = (N_{11}/N) \times \log_2(N \times N_{11} / (N_{1.} \times N_{.1})) + (N_{01}/N) \times \log_2(N \times N_{01} / (N_{0.} \times N_{.1})) + (N_{10}/N) \times \log_2(N \times N_{10} / (N_{1.} \times N_{.0})) + (N_{00}/N) \times \log_2(N \times N_{00} / (N_{0.} \times N_{.0}))$$

Fully Solved MI Example (from Slides — Reuters-RCV1 Dataset)

Class = 'poultry' | Term = 'export' | Total documents $N = 801,948$

	ec = poultry (class=1)	ec = NOT poultry (class=0)
et = export (term=1)	$N_{11} = 49$	$N_{10} = 27,652$
et = NOT export (term=0)	$N_{01} = 141$	$N_{00} = 774,106$

Step 1 — Calculate marginal totals:

$$\begin{aligned} N_{1.} &= N_{11} + N_{10} = 49 + 27,652 = 27,701 \text{ (docs containing 'export')} \\ N_{0.} &= N_{01} + N_{00} = 141 + 774,106 = 774,247 \text{ (docs NOT containing 'export')} \\ N_{.1} &= N_{11} + N_{01} = 49 + 141 = 190 \text{ (docs in class 'poultry')} \\ N_{.0} &= N_{10} + N_{00} = 27,652 + 774,106 = 801,758 \text{ (docs NOT in class 'poultry')} \end{aligned}$$

Step 2 — Plug into MI formula:

$$\begin{aligned} \text{Term 1 (} N_{11} \text{): } &(49/801948) \times \log_2(801948 \times 49 / (27701 \times 190)) \\ \text{Term 2 (} N_{01} \text{): } &(141/801948) \times \log_2(801948 \times 141 / (774247 \times 190)) \\ \text{Term 3 (} N_{10} \text{): } &(27652/801948) \times \log_2(801948 \times 27652 / (27701 \times 801758)) \\ \text{Term 4 (} N_{00} \text{): } &(774106/801948) \times \log_2(801948 \times 774106 / (774247 \times 801758)) \end{aligned}$$

Step 3 — Final Result:

$$I(U;C) \text{ for 'export' in class 'poultry'} \approx 0.0001105$$

This is a SMALL MI value — meaning 'export' is not very informative for the 'poultry' class. Compare: the word 'poultry' itself would have a MUCH higher MI score for this class!

MI Example Table from Slides (High-Scoring Terms per Class)

Class: UK	Score	Class: China	Score	Class: Sports	Score
london	0.1925	china	0.0997	soccer	0.0681
uk	0.0755	chinese	0.0523	cup	0.0515
british	0.0596	beijing	0.0444	match	0.0441
stg	0.0555	yuan	0.0344	matches	0.0408
britain	0.0469	shanghai	0.0292	played	0.0388

Observation: Terms with HIGH MI are class-specific (e.g., 'london' for UK, 'soccer' for Sports). These are exactly the features that should be kept!

Topic 9: Chi-Square Method (X2)

What Is It?

Chi-Square measures how much the EXPECTED counts and OBSERVED counts of a term-class combination deviate from each other. A HIGH X2 value means the term and class are NOT independent — the term strongly relates to the class and should be kept as a feature.

The Formula:

$$X2(D, t, c) = (N11+N10+N01+N00) \times (N11*N00 - N10*N01)^2 / [(N11+N01) \times (N11+N10) \times (N10+N00) \times (N01+N00)]$$

Where: N11, N10, N01, N00 are the same four contingency counts as in MI. A high X2 value = hypothesis of independence is INCORRECT = term IS related to class = KEEP IT. A low X2 value = term and class are independent = term is NOT useful = REMOVE IT.

Fully Solved Chi-Square Example (Same Reuters Data)

Class = 'poultry' | Term = 'export' | N = 801,948

	ec = poultry (class=1)	ec = NOT poultry (class=0)
et = export (term=1)	N11 = 49	N10 = 27,652
et = NOT export (term=0)	N01 = 141	N00 = 774,106

Step 1 — Compute Expected Value E11:

$$E11 = N \times P(t) \times P(c) = N \times (N11+N10)/N \times (N11+N01)/N = 801,948 \times (49+27652)/801948 \times (49+141)/801948 = 801,948 \times (27701/801948) \times (190/801948) \approx 6.6$$

Step 2 — Compute ALL Four Expected Values (Same Logic):

General Formula: $E(et, ec) = N \times P(\text{term}=et) \times P(\text{class}=ec)$ Each E value = Total docs x (row total / N) x (column total / N) = (row total x column total) / N

E11 — term present AND class present:

$$E11 = (N11+N10) \times (N11+N01) / N = (\text{row1 total}) \times (\text{col1 total}) / N = (49+27,652) \times (49+141) / 801,948 = 27,701 \times 190 / 801,948 = 5,263,190 / 801,948 \approx 6.6$$

E10 — term present BUT class NOT present:

$$E10 = (N11+N10) \times (N10+N00) / N = (\text{row1 total}) \times (\text{col0 total}) / N = (49+27,652) \times (27,652+774,106) / 801,948 = 27,701 \times 801,758 / 801,948 = 22,209,542,558 / 801,948 \approx 27,694.4$$

E01 — term NOT present BUT class present:

$$E01 = (N01+N00) \times (N11+N01) / N = (\text{row0 total}) \times (\text{col1 total}) / N = (141+774,106) \times (49+141) / 801,948 = 774,247 \times 190 / 801,948 = 147,106,930 / 801,948 \approx 183.4$$

E00 — term NOT present AND class NOT present:

$$E00 = (N01+N00) \times (N10+N00) / N = (\text{row0 total}) \times (\text{col0 total}) / N = (141+774,106) \times (27,652+774,106) / 801,948 = 774,247 \times 801,758 / 801,948 \approx 774,063.6$$

Summary of All Expected Values:

Cell	Row Total	Col Total	Calculation	Expected E
E11	27,701	190	$27,701 \times 190 / 801,948$	≈ 6.6
E10	27,701	801,758	$27,701 \times 801,758 / 801,948$	$\approx 27,694.4$
E01	774,247	190	$774,247 \times 190 / 801,948$	≈ 183.4
E00	774,247	801,758	$774,247 \times 801,758 / 801,948$	$\approx 774,063.6$

Quick Check: $E11+E10+E01+E00 = 6.6+27694.4+183.4+774063.6 = 801,948 = N$ (must always equal N!)

Step 3 — Plug into simplified X2 formula:

$$\begin{aligned} \text{Numerator} &= (N11+N10+N01+N00) \times (N11 \times N00 - N10 \times N01)^2 = 801,948 \times (49 \times 774,106 - 27,652 \times 141)^2 = 801,948 \times (37,931,194 - 3,898,932)^2 = 801,948 \times (34,032,262)^2 \\ \text{Denominator} &= (N11+N01) \times (N11+N10) \times (N10+N00) \times (N01+N00) = 190 \times 27,701 \times 801,758 \times 774,247 \end{aligned}$$

$$X2(D, t, c) \approx 284$$

X2 = 284 is a HIGH value! This means 'export' and 'poultry' are NOT independent — 'export' IS a useful feature for classifying 'poultry' documents. Compare with the MI value of 0.0001105 which seemed very small but still indicated some relationship.

MI vs Chi-Square — Key Comparison

	Mutual Information	Chi-Square
Measures	How much term reduces uncertainty	Deviation of observed from expected
Scale	Very small values (0 to ~0.2)	Larger values (0 to hundreds+)
Bias	Biased toward RARE terms	Biased toward FREQUENT terms
Interpretation	Higher = more informative	Higher = stronger term-class relationship
Both agree?	Usually yes for top features	Can differ for mid-range features

Topic 10: Frequency-Based Feature Selection

What Is It?

The simplest feature selection method — just select the terms that appear MOST FREQUENTLY in each class. No complex formula needed.

Two Types of Frequency:

Type	Definition	Best for
Document Frequency	Number of DOCUMENTS in class c that contain term t	Bernoulli model
Collection Frequency	Number of TOKENS of term t that occur in all documents in class c	Multinomial model

Advantage and Disadvantage

Advantage	Disadvantage
Extremely simple and fast to compute	Selects common terms that may have NO class-specific info
No complex formula required	Stop words like "the", "a", "is" can score highly
Works reasonably well in practice	Less precise than MI or Chi-Square

Comparison of All Three Feature Selection Methods

Method	Formula	What it captures	Strength	Weakness
Mutual Information	$I(U_t; C_c)$	Term-class information gain	Theoretically principled	Biased toward rare terms
Chi-Square	$X^2(t, c)$	Observed vs expected deviation	Handles frequent terms well	Biased toward frequent terms
Frequency	$N(t, c)$	Raw term count in class	Simple and fast	May select non-informative terms

From the Feature Selection Experiment graph in slides: Both MI and Chi-Square significantly outperform frequency-based selection, especially at smaller feature set sizes. MI and Chi-Square perform similarly overall, with Chi-Square slightly better for the multinomial model on the Reuters dataset.